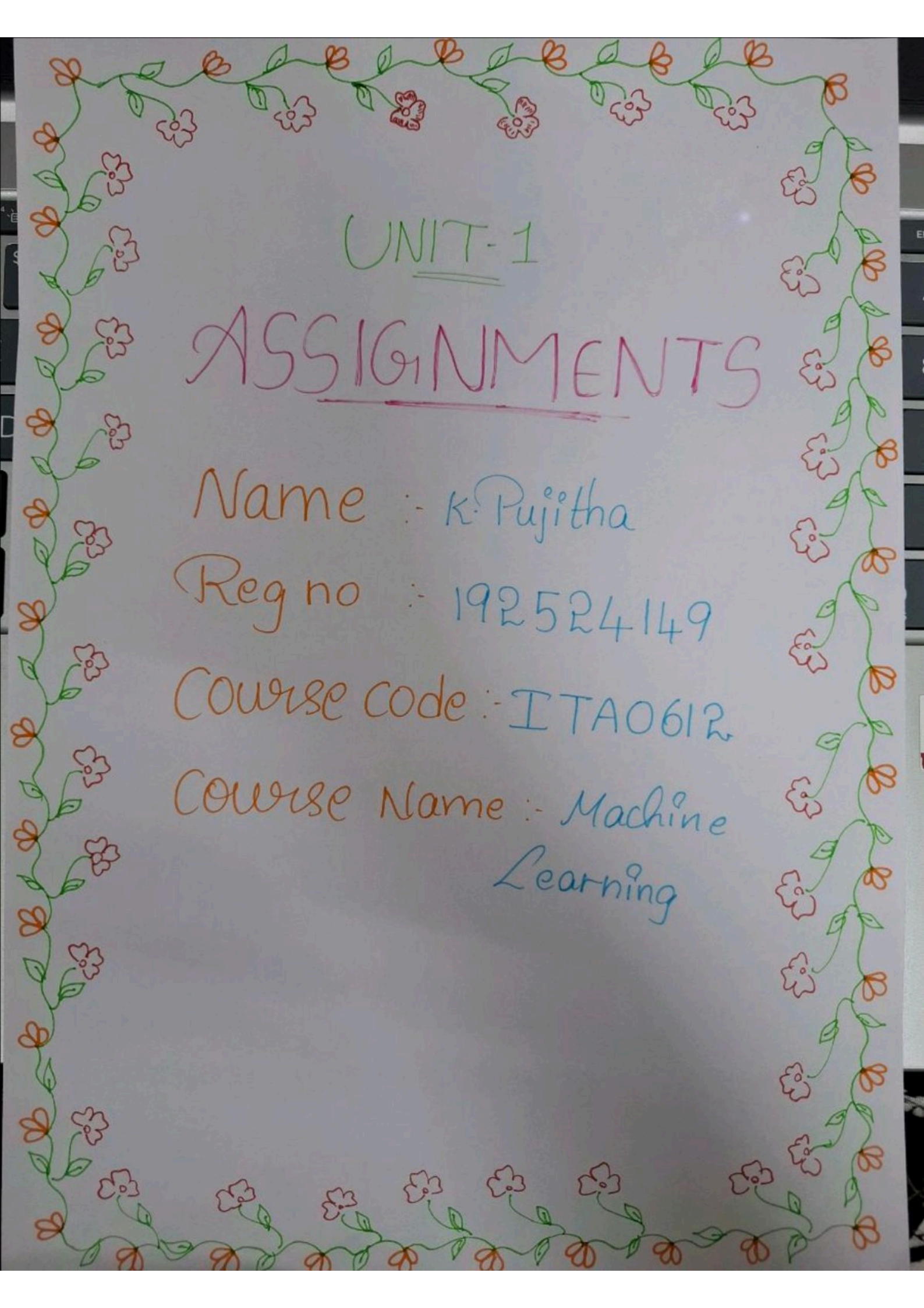




UNIT-1

ASSIGNMENTS

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Course code :- ITA0612

Course Name :- Machine
Learning

Unit-1

Analytical Assignments with SDGs

Machine learning

①

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- ① Explain how machine learning can identify poverty-prone regions using socio-economic data.

A. ML to identify Poverty-Prone Regions

Machine learning can identify areas that are highly affected by poverty by analyzing socio-economic data.

Steps:-

- ① Collect data such as income, employment, Education, housing, access to basic facilities.
- ② Preprocess the data by handling missing values and removing incorrect data.
- ③ Select important features such as income level, Education level.
- ④ Train an ML model such as Decision Tree.
- ⑤ The model classifies regions as high, medium, and risk areas.

Benefit:

Governments can identify poverty-prone regions and provide targeted financial assistance, Education, health care.

- ② Design a Machine learning model to identify Eligible beneficiaries for Government welfare schemes.

A. ML to identify Eligible Beneficiaries for welfare schemes

Machine learning can help governments identify people who are Eligible for welfare schemes.

Steps:

- ① Collect data such as income, family size, Employment, age, disability & living conditions.
- ② Clean and preprocess the data.
- ③ Select important Eligibility-related features.
- ④ Train a classification model such as Decision Tree, Random forest or logistic Regression.
- ⑤ The model predicts whether a person is eligible or not.

Benefit:-

It reduces manual errors, improves transparency, Prevents duplicate claims, and helps benefits reach the correct People.

(2)

2) Explain how machine learning can predict crop yield using historical agricultural data.

ML to Predict Crop Yield - SDG 2: zero Hunger

A. Important data

- Rainfall
- Temperature
- Soil quality
- Fertilizer usage
- Irrigation
- Previous crop yield

Steps:

- 1) Collect historical agricultural data
- 2) Clean and preprocess the data
- 3) Select important features.
- 4) Train a Regression model such as Linear Regression, Random forest.
- 5) Predict the expected crop yield for particular season.

Benefit:

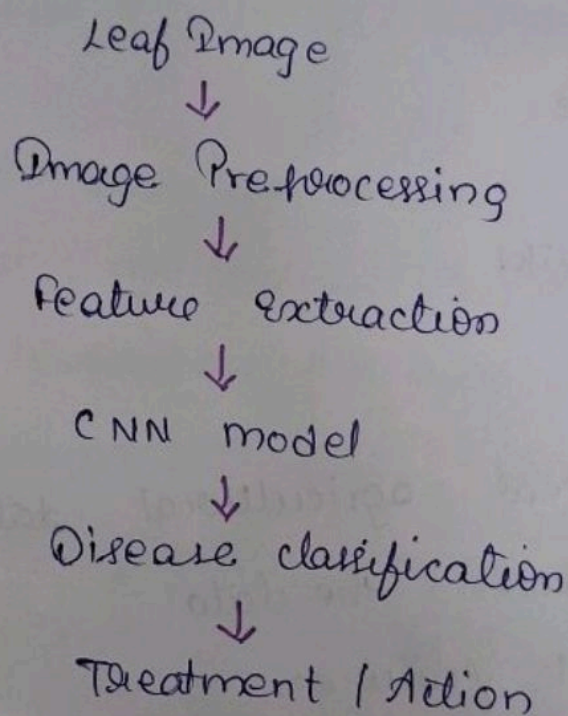
Formers can plan irrigation, fertilizers and resources properly, improving productivity and shortages.

- 4) Develop a Machine learning solution for crop disease detection using leaf images.

1. ML for crop disease Detection using leaf Images

Machine learning can automatically identify crop diseases from photographs of leaves.

steps



- 1) Collect healthy and diseased leaf images.
- 2) Resize and normalize the images.
- 3) Train a Convolutional Neural Network (CNN) using labelled images.
- 4) The model learns visual symptoms such as spots, discoloration & damaged areas.
- 5) For a new leaf image, the model predicts the disease.

Q. Explain how machine learning can reduce food wastage in supermarkets.

A. ML to learning Reduce food wastage in supermarkets :-

Machine learning can predict product demand and help supermarkets manage their inventory.
steps:

- 1) Collect historical sales, seasonal, Customer-demand, and Expiry-date data.
- 2) Preprocess and analyze the data.
- 3) Train on ML model to predict future demand
- 4) maintain stock according to predicted demand.
- 5) Products approaching expiry can be given discounts or promotions.

Historical sales Data



ML Prediction



Future Demand



Better Inventory management



Reduced Food waste